OUTLAND ADVENTURES

DATABASE CASE STUDY

REBECCA ESSENBURG
&
DANIEL FRYER

CASE INTRODUCTION

This case study was about two people who enjoyed the outdoors and decided to open a business of taking people on outdoor adventures around the world. They have two guides, a marketing director, a supply manager, and an ecommerce developer. Their goal was to find ways to optimize their operations. There were three questions they wanted answered:

1. Do enough customers buy equipment to keep equipment sales?

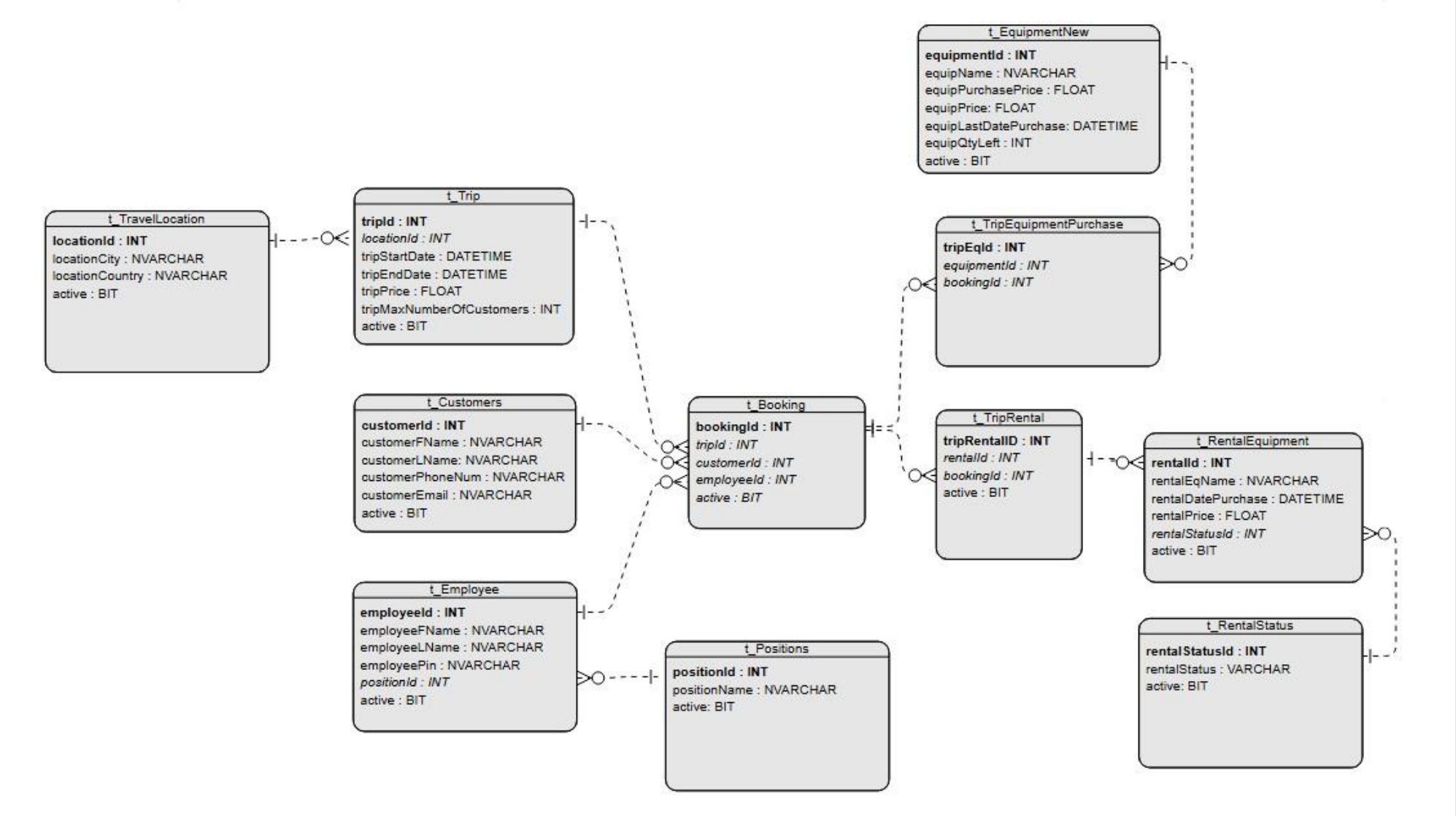
2. Are there any inventory items that are over five years old?

3. Is there any one region (Africa, Asia, or Southern Europe) that has a downward trend in bookings?

GROUP 4’S APPROACH

For this case study, we started from the heart of the information – the trips themselves – and branched outward to include as much business-relevant data as we could within our system. This database both answers the owner’s pressing questions while retaining important company and customer information.

ENTITY RELATIONSHIP DIAGRAM



DO ENOUGH CUSTOMERS BUY EQUIPMENT TO KEEP EQUIPMENT SALES?

--Customers Buying Equipment--

Customer	Total Purchases	Total Sales
Mark Hanson	4	677.96
Billy Bob	1	125.99
Jill Hanson	1	125.99

--Equipment Inventory Purchased--

Item	Total Items Purchased	Total Sales Amount
2 Person Tent	1	249.99
First Aid Kit	4	503.96
Camp Stove	1	175.99

The first report allows the business owners to see which customers have purchased (not rented) equipment for their trips. They can see customers by name, how many items each customer has purchased, as well as the total amount spent on these items to make their decision on whether equipment sales are providing enough profit overall. The second report is more optional, but it allows the business owners to see how much of each individual item was purchased if they would like to see which inventory is the least popular.

ARE THERE ANY INVENTORY ITEMS THAT ARE OVER FIVE YEARS OLD?

```
--Rental Equipment > 5 Years Old--
+----------+------------------+-------------------------+----------------+
|   Item ID | Equipment        | Date Added to Inventory | Rental Status  |
+==========+==================+=========================+================+
|         2 | Hydration Bladder | 2020-01-01 00:00:00     | Booked         |
+----------+------------------+-------------------------+----------------+
|         6 | Sleeping Bag     | 2020-07-07 00:00:00     | Last Use       |
+----------+------------------+-------------------------+----------------+

--New Equipment > 5 Years Old--
+----------+------------+-------------------------+
|   Item ID | Equipment  | Date Added to Inventory |
+==========+============+=========================+
|         4 | Camp Stove | 2020-11-03 00:00:00     |
+----------+------------+-------------------------+
```

There are two reports to answer this question, one which displays rentable equipment and one which displays new equipment. These tables only display items that have been in inventory for over 5 years from the current date, as well as their item ID to distinguish that specific item from others of its kind for easy removal/inspection. It will not display any items newer than 5 years old.

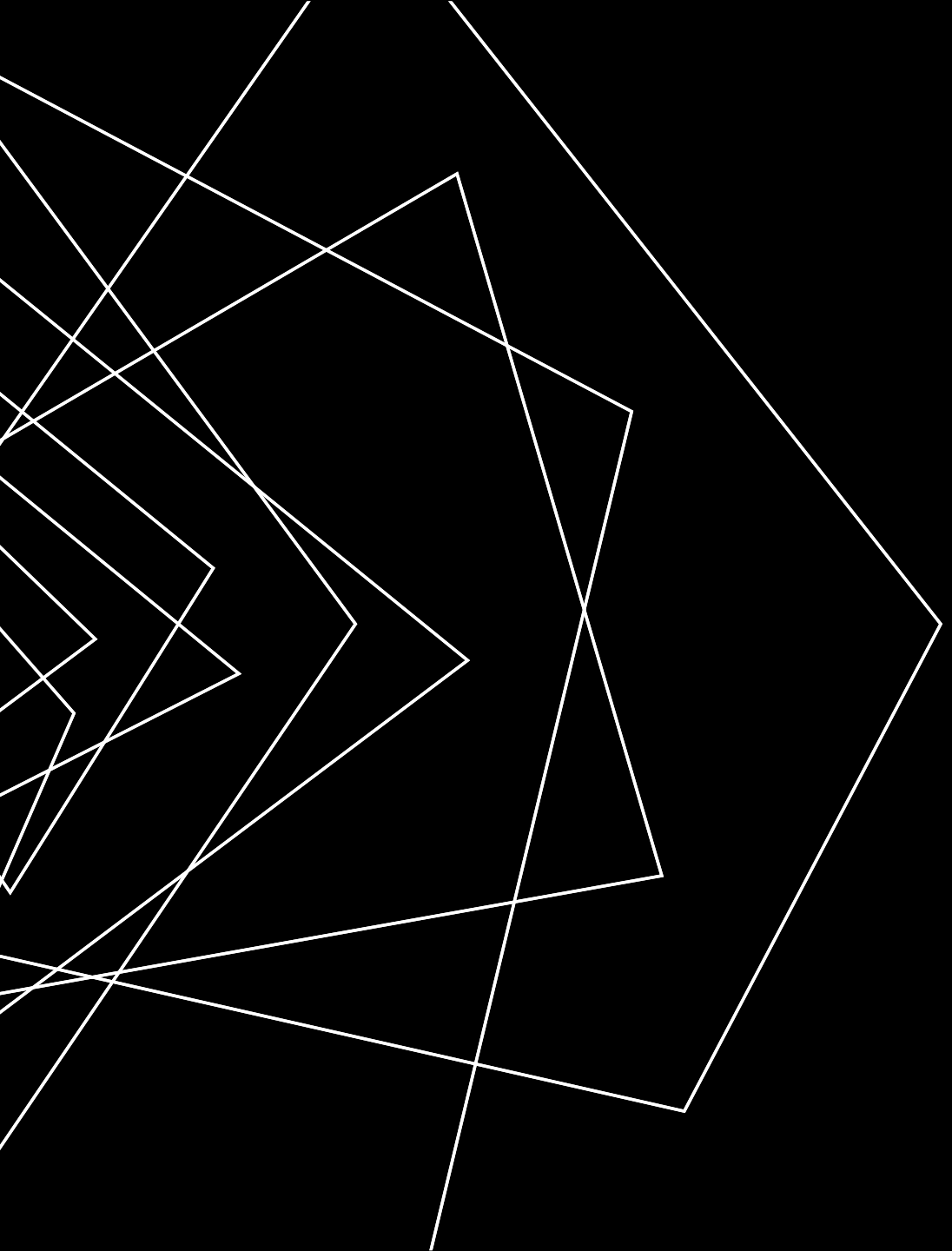
IS THERE ANY ONE REGION (AFRICA, ASIA, OR SOUTHERN EUROPE) THAT HAS A DOWNWARD TREND IN BOOKINGS?

```
--Bookings by Region--
+----------+----------------+
| Region   |   TotalBookings |
+==========+================+
| Africa   |              6 |
+----------+----------------+
```

This question can be answered by a single report. Each booking has been labeled by which region it was in, and the total number of bookings are added together for each region. In our example database, all the bookings are within the Africa region.

CASE ASSUMPTIONS

- Customers would have to register before booking. No guest accounts.
- Purchased equipment can’t be used as rental and rental equipment can’t be sold as new equipment.
- A person can purchase and/or rent equipment for each trip.
- Trips don’t include equipment and people would have to purchase or rent the equipment.
- Only a maximum numbers of customers can be signed up for each trip.
- An employee can only be on one trip at a time.



THANK YOU